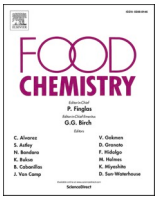
Food Chemistry 405 (2023) 134828



Contents lists available at ScienceDirect

Food Chemistry

journal homepage: www.elsevier.com/locate/foodchem





Rapid detection of adulteration of olive oil with soybean oil combined with chemometrics by Fourier transform infrared, visible-near-infrared and excitation-emission matrix fluorescence spectroscopy: A comparative study

Xiangru Meng, Chunling Yin *, Libo Yuan, Yan Zhang, Ying Ju, Kehui Xin, Wenbo Chen, Kaidi Lv, Leqian Hu *

School of Chemistry and Chemical Engineering, Henan University of Technology, Zhengzhou 450001, China

ARTICLE INFO

Keywords:
Olive oil
Spectroscopic techniques
Adulteration
Pattern recognition
Comparative study

ABSTRACT

Several spectroscopic techniques have been used to detect olive oil adulteration. To evaluate the performance of these spectral techniques on this issue, this work performed a comparative study on identifying adulterated olive oil with different concentrations of soybean oil based on Fourier-transform infrared (FTIR), visible-near-infrared (Vis-NIR) and excitation-emission matrix fluorescence spectroscopy (EEMs) combined with chemometrics. Principal component analysis (PCA)/ multi-way-PCA analysis showed the feasibility of the three spectral methods for the identification of olive oil adulteration. The accuracy of FTIR and Vis-NIR based on partial least squares discriminant analysis (PLS-DA) for adulterated olive oil was 100%, while the accuracy of EEMs based on unfold-PLS-DA was only 73%. The accuracy of EEMs combined with back-propagation artificial neural network based on self-weighted alternating trilinear decomposition is 100%. In comparison, FTIR and Vis-NIR are superior for the detection of olive oil adulteration due to the convenience of instrument operation and modeling.

1. Introduction

Olive oil is a kind of cold-pressed vegetable oil extracted from olive fruits. It is rich in unsaturated fatty acids (main oleic acid), and also contains a variety of valuable bioactive small molecules such as tocopherols, polyphenols (hydroxytyrosol, tyrosol, and oleuropein, etc.), squalene, sterols and β-carotene (Uncu & Ozen, 2020). It has been found that olive oil could not only increase the taste of food but also reduce blood lipids, prevent cardiovascular diseases, and have a good protective effect on human health (Bermudez et al., 2011; Romani et al., 2019; Bucciantini, Leri, Nardiello, Casamenti, & Stefani, 2021; Stark & Madar, 2002). Due to its high nutritional value, olive oil is more expensive than most other edible oils. Driven by economic interests and the discrepancy between supply and demand, some unscrupulous merchants adulterate olive oil for profit, and the most typical olive oil fraud is to blend lower-priced vegetable oils (palm oil, soybean oil, etc.) into olive oil (Li, Chen, Chen, Guo, Li, & Xu, 2020; Vieira, Assis, de Queiroz, Neves, & de Oliveira, 2021). These adulterated oils infringe on the rights of consumers, threaten the health of human, violate the law, disrupt the market, and challenge the quality inspection of olive oil.

Over the years, several analysis techniques and methods have been studied on identifying adulteration in olive oil, including sensory analysis, classical chemical analysis, chromatography, spectroscopy, electronic nose, electronic tongue, and DNA method. Among them, the sensory evaluation method is easy to be affected by inspectors, environment, evaluation criteria, and other factors. It has low accuracy, low repeatability, and high labor cost, thus limiting the application of olive oil adulteration detection. Classical chemical analysis and chromatographic techniques (Carranco, Farres-Cebrian, Saurina, & Nunez, 2018; Monfreda, Gobbi, & Grippa, 2014) are based on the quantitative detection of single or multiple specific chemical compositions (acid value, peroxide value, fatty acid, tocopherol, etc.), so they are difficult to obtain complete information of oil samples. Although electronic nose and electronic tongue have been developed for the detection of olive oil adulteration (Harzalli et al., 2018; Zarezadeh, Aboonajmi, Varnamkhasti, & Azarikia, 2021), their application is limited due to poor selectivity, low sensitivity, and signal noise. DNA technology has been developed to identify adulteration of olive oil effectively, because it can provide information on the characteristics of cultivars (botanical) (Vietina, Agrimonti, & Marmiroli, 2013). However, complex sample

* Corresponding authors.
E-mail addresses: clyin@yeah.net (C. Yin), leqianhu@163.com (L. Hu).

https://doi.org/10.1016/j.foodchem.2022.134828
Received 18 May 2022; Received in revised form 23 October 2022; Accepted 30 October 2022
Available online 4 November 2022
0308-8146/© 2022 Elsevier Ltd. All rights reserved.

pretreatments need to be done and toxic solvents are used in the process for these analytical techniques. It is time-consuming and not environmentally friendly.

Currently, spectroscopy methods (Uncu & Ozen, 2019; Vieira, Assis, de Queiroz, Neves, & de Oliveira, 2021; Omwange et al., 2021; Meras, Manzano, Rodriguez, & de la Pena, 2018; de Lima, Musso, & Bertoldo Menezes, 2020; Hou et al., 2021) can overcome shortcomings of the above methods, and exhibit advantages of low operating cost, fast analysis speed, high accuracy and sensitivity, non-destructive, and no sample preparation process. Furthermore, spectral techniques can provide multiple chemical information about olive oil as they are mainly based on spectral fingerprint information rather than being limited to a particular component. At the same time, combined with chemometrics, a large number of spectral data can be analyzed by mathematical separation. Therefore, spectroscopy has become increasingly popular in the detection of olive oil adulteration, especially mid-infrared (MIR), near-infrared (NIR) and fluorescence spectroscopy techniques. Among them, MIR spectrum and NIR spectrum belong to the infrared spectrum. The difference is that NIR records the overtone and combined band information of the fundamental vibrations of a single chemical bond in a molecule, while MIR records the fundamental absorptions of the vibrations of bonds in a molecule. The range of NIR and MIR analysis can cover almost all organic compounds in olive oil, and they give information regarding the composition, structure, and interaction between different functional groups in olive oil. In addition, three-dimensional fluorescence spectroscopy (also known as excitation-emission matrix fluorescence spectroscopy, EEMs) combined with chemometrics can process complex overlapping fluorescence signals and obtain comprehensive fluorescence information of olive oil in a short time, which has been successfully used in the detection of oil adulteration (Omwange et al., 2021; Meras, Manzano, Rodriguez, & de la Pena, 2018).

Although spectroscopic methods have been successfully applied, it is very important to investigate the capabilities of these techniques for rapid and accurate identification of olive oil adulteration. However, most of these reported studies have only used one specific spectral technique, and only a few reports have compared the performance of different spectral methods in the identification of olive oil adulteration. To evaluate the performance of common spectral techniques on this issue, it is necessary to make a comparative study on the identification of adulteration in olive oil. Therefore, this study aimed to compare the ability of MIR, NIR and EEMs combined with pattern recognition methods to rapidly detect different levels of adulterated olive oil.

2. Materials and methods

2.1. Sample collection

Edible oil samples, including extra virgin olive oil (Luhua brand, 30 different batches, China, the origin of olives is Spain) and soybean oil (Yingma brand, 3 different batches, Zhongshan, Guangdong Province, China), were purchased from supermarkets in Zhengzhou, Henan Province, China. All the oil samples were stored in the dark at room temperature until analysis day. Ten olive oil samples and one soybean oil sample were selected at random to form a group. Soybean oil was used as an adulterant. Adulterated oil samples were prepared by adding amounts of soybean oil (5 %, 10 %, 20 %, 30 %, and 50 %, w/w) to the pure olive oil samples. Thus, 50 samples of adulterated oil were obtained. Each adulterated oil sample was mixed by vortex for 3 min at room temperature. The rest of oil samples were divided into two groups by the same method to prepare adulterated oil samples. As a result, a total of 180 samples were collected in this study, including 150 adulterated oil samples (30 samples for each adulteration level) in different concentrations and 30 pure olive oil samples, which were used to build the adulteration identification model for olive oil adulterated with soybean oil.

2.2. Instrumentation

2.2.1. Fourier-transform infrared (FTIR) spectroscopy measurements

Mid-infrared spectra between 4000 and 400 cm^{-1} of oil samples were recorded using PerkinElmer Spectrum TWO FTIR spectrometer (PerkinElmer, Shelton, CT, USA) equipped with a deuterated triglycine sulfate detector (DTGS). As a sampling technique, attenuated total reflectance (ATR) accessory with ZnSe diamond crystal was used. FTIR information was collected by taking a 5 μL edible oil sample and dropping the oil sample on the crystal's surface attached to the ATR. After each analysis, the crystal for sampling was carefully cleaned by hexane and anhydrous ethanol, respectively, for three times. In a dry environment at room temperature, the FTIR spectral data of each olive oil sample were collected with air as the detection background. The spectral scanning range was 4000–400 cm^{-1}, the resolution was 8 cm^{-1}, and the scanning times were 16. The spectra were recorded as absorbance (A) values at each data point. Spectral scans of each sample were taken in triplicate. Subsequently, the results were reported as mean values for statistical processing. A blank background was documented after every three scans.

2.2.2. Visible-near-infrared (Vis-NIR) spectroscopy measurements

Visible-near-infrared spectra between 400 and 2500 nm (spectral bandpass 10 nm ± 1 nm) of edible oil samples were recorded by using Foss XDS Rapid Content Analyzer (Foss, Hillerød, Denmark) equipped with a spinning module. All measurements were taken in a dry environment at 40 °C. Air was the detection background, and the spectral data of an appropriate amount of oil samples into the sample tube with a diameter of 6.0 mm were collected. After each analysis, the sample tube was cleaned using hexane and anhydrous ethanol, respectively, for three times. The spectral scanning range was 400–2500 nm, the resolution was 2 nm, and the scanning times was 32. The spectra were recorded as absorbance (A) values at each data point. Spectral scans of each sample were taken in triplicate, and the results were reported as mean values. A blank background was documented after every three scans.

2.2.3. Excitation-emission fluorescence spectroscopy measurements

Excitation-emission fluorescence spectroscopy measurement of olive oil samples were performed using a Hitachi F-7000 fluorescence spectrometer (Hitachi, Beijing, China) equipped with a 150 W xenon lamp. Each olive oil sample was scanned using a 3 mL quartz cell (a light path of 1 cm) by fluorescence spectrometer, and the corresponding EEMs were obtained at room temperature. Each EEMs consisted of measuring excitation wavelengths from 280 to 710 nm (2 nm intervals) and emission wavelengths from 300 to 730 nm (2 nm intervals). Thus 216 emission spectral data points and 216 excitation spectral data points were obtained, and its dimensions is 216 × 216. These matrices were represented in the form of contour maps (the horizontal axis represents the emission wavelength, the vertical axis represents the excitation wavelength, and the contour line represents the fluorescence intensity). After each measurement, the cuvette was cleaned using hexane and anhydrous ethanol respectively, for three times, to dry and eliminate any remaining fat. Furthermore, other fluorescence measurement parameters were as follows: the scan speed was set as 2400 nm/min, the detector voltage was 550 V, and the slit width was adjusted to 5 nm for both excitation and emission. A blank background was scanned after every three samples.

2.2.4. Chemometric models and software

Useful information about the molecular composition, structure, and interactions within a sample can be provided by analyzing the results of Vis-NIR, FTIR and EEMs. These methods are relatively simple, non-destructive to the analyzed samples, reproducible, do not require sample preparation, and avoid using a large amount of solvent (Gendrin, Roggo, & Collet, 2008). However, since the main ingredients in edible oil are similar (mainly saturated fatty acid, oleic acid, linoleic acid, and

so on). The signals from different adulterated olive oil samples are similar and rather complex, which make them difficult to interpret. Hence, it is still difficult to distinguish the adulterated olive oil accurately by directly using their spectroscopy profiles. Meanwhile, the detection data of olive oil samples from various spectroscopic techniques are enormous. Thus, it is necessary to distinguish adulterated olive oil by chemometrics methods.

Pattern recognition techniques in chemometrics are the most commonly used methods for edible oil adulteration analysis. These techniques can be further classified as unsupervised pattern recognition like principal component analysis (PCA) (Amit, Jamwal, Kumari, Dhaulaniya, Balan, & Singh, 2020) and cluster analysis as well as supervised pattern recognition such as soft independent modeling of class analogy (SIMCA) (Monfreda, Gobbi, & Grippa, 2014), linear discriminant analysis (LDA) (Li, Chen, Chen, Guo, Li, & Xu, 2020), partial least squares discriminant analysis (PLS-DA) (Vieira, Assis, de Queiroz, Neves, & de Oliveira, 2021), support vector machine (SVM) (Omwange et al., 2021), canonical correlation analysis (CCA) and so on (Rohman, 2017). In this study, the analysis software Matlab 2012a (MathWorks, Natick, MA, USA) was used for spectral data processing to distinguish olive oil adulterated with soybean oil.

For comparative and practical purposes, it is more reasonable to choose the same and common pattern recognition methods to identify the adulteration of edible oil. Thus, PCA and PLS-DA were used to construct the model in this work. For EEMs data, traditional 2D pattern recognition methods such as PCA, SIMCA and PLS-DA will no longer be suitable for classifying the EEMs datasets. Multidimensional pattern recognition methods, i.e., multi-way principal component analysis (M-PCA, an extension of PCA for three-dimensional data) (Pardo, Vega, Barrado, Castrillejo, & Sanchez, 2013) and unfold partial least squares

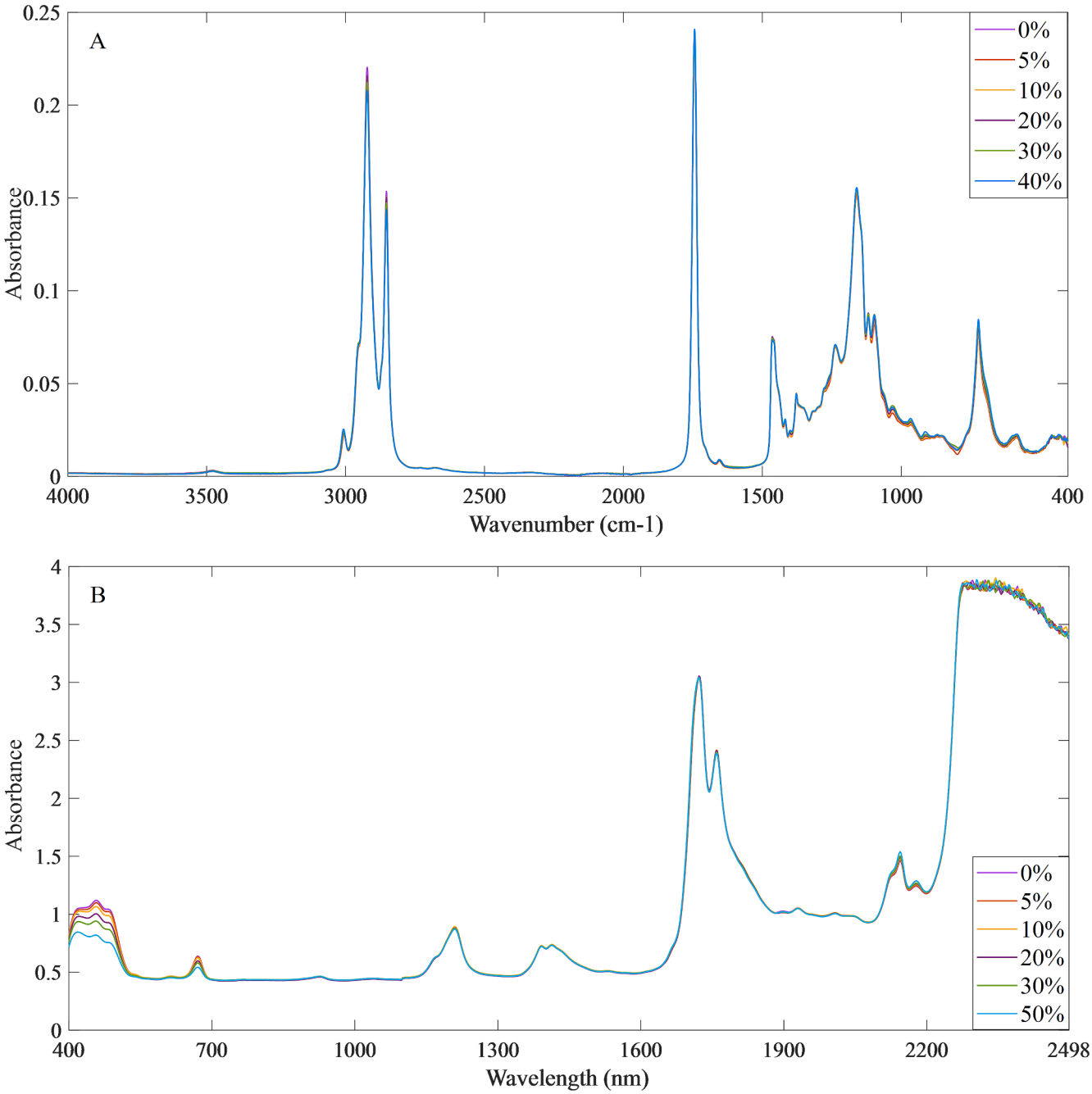


Fig. 1. FTIR (A) and Vis-NIR (B) spectra for adulterated olive oil samples.

discriminant analysis (U-PLS-DA, an extension of PLS-DA for higher orders data) (da Silva, Soares, Insausti, Galvao, Band, & de Araujo, 2016) were exploited to build models because of three-dimensional characteristics of fluorescence fingerprint data. In the last, self-weighted alternating trilinear decomposition (SWATLD) (Chen, Wu, Jiang, Li, & Yu, 2000; Chen, Wu, & Yu, 2001) combined with back-propagation artificial neural network (BP-ANN) (Rumelhart, Hinton, & Williams, 1986; Deng et al., 2021) was also used to identify adulterated olive oil.

3. Results and discussion

3.1. FTIR, Vis-NIR and EEMs spectra of olive oil samples

The typical raw FTIR spectra of pure olive oil samples and adulterated olive oil samples with different levels of adulteration at wavenumber regions of 4000–400 cm^{-1} are shown in Fig. 1A. As depicted by the FTIR spectra of samples, the sample has characteristic absorptions in different bands such as 3007, 2923, 2853, 1743, 1464, 1379, 1237, 1163, 1119, 1099, and 723 cm^{-1}. The absorbance at 3007 cm^{-1} is relevant to the C—H stretching vibration of –HC = CH–(*cis*) groups. The absorbance at 2923 cm^{-1} and 2853 cm^{-1} are related to asymmetrical

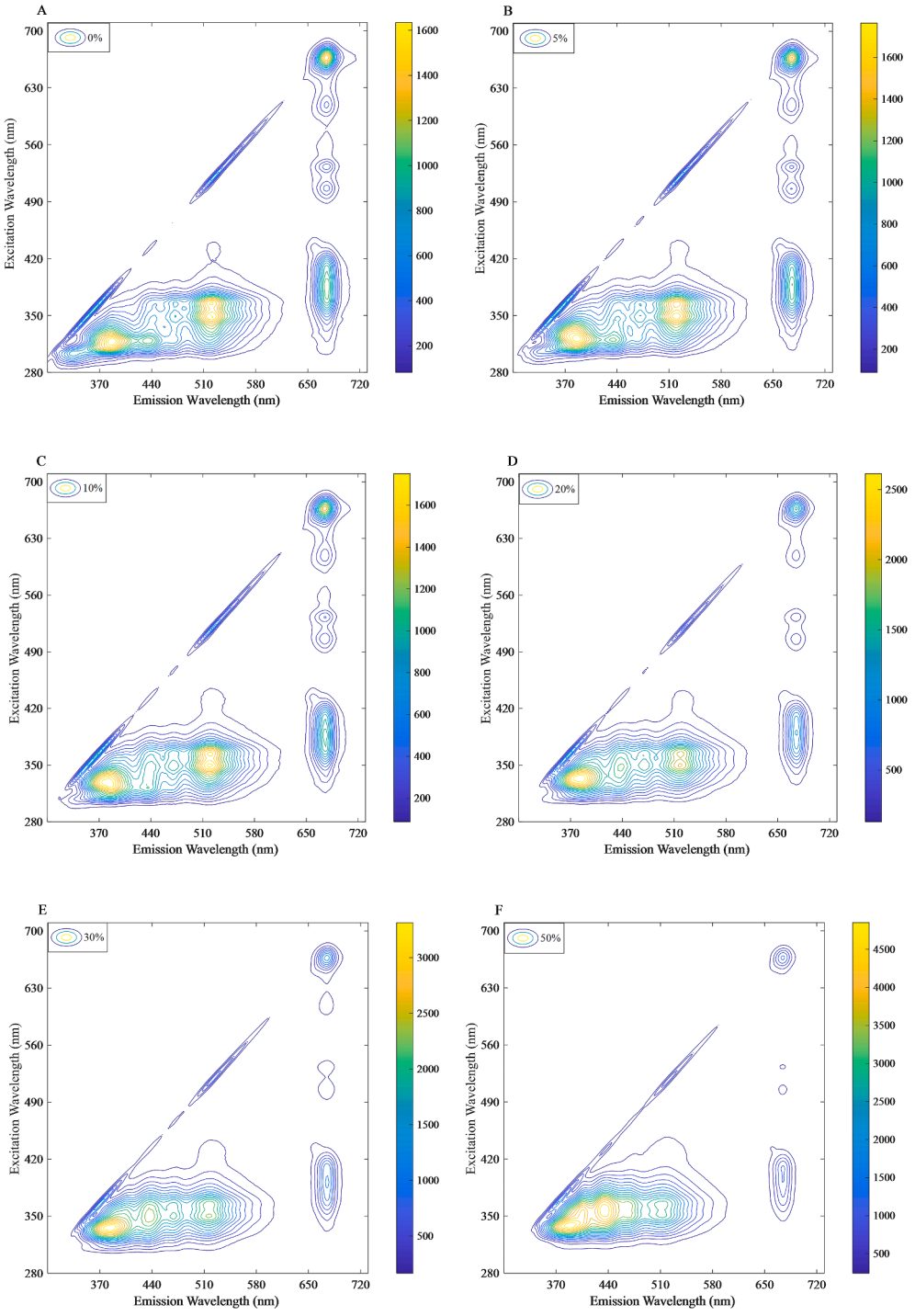


Fig. 2. Contour plots of raw fluorescence fingerprints spectra (EEMs) of adulterated olive oil samples.

and symmetrical stretching vibrations of aliphatic –CH_2– and –CH_3 groups, respectively. The major peak at 1743 cm^{-1} is associated with the stretching vibration of –C═O of triglycerides. The major bands at 1464 cm^{-1} and 1379 cm^{-1} are associated with aliphatic –CH_2– and –CH_3 scissor vibrations, respectively. The absorptions at 1237 cm^{-1} and 1163 cm^{-1} are related to the stretching vibration of C—O and the bending vibration of –CH_2– in the aliphatic ester group, respectively. The remaining peaks at 1119 cm^{-1} and 1099 cm^{-1} are correlated with C—O stretching vibration, while a peak at 723 cm^{-1} is associated with the bending rocking vibration of –CH_2– (Amit et al., 2020).

Vis-NIR spectra of pure and adulterated olive oil samples are presented in Fig. 1B. It shows that the original spectral profiles of all olive oil samples are very similar, only with slight differences in absorption intensity. In the near-infrared spectra wavelength range, there is a large amount of absorption at 1100–2300 nm, and the main absorption peaks are distributed in the regions of 1100–1250 nm, 1350–1500 nm, 1700–1900 nm, and 2100–2200 nm. This points to the level variation among free fatty acid, unsaturated degree, *trans*-fatty acid and lipid oxidation (Du et al., 2021). The absorption peak at 1210 nm is related to the second overtones of the C—H bonds stretching vibration from –CH_3 and –CH_2– groups in aliphatic hydrocarbon, while peaks at 1722 nm and 1758 nm are due to the first overtone of C—H from –CH_2– groups. The absorption bands in 1392 nm and 1414 nm are from the combinations of vibrations of C—H bonds of –CH_2– and –CH_3 groups in a straight alkane. The bands from 2120 nm to 2176 nm may be attributed to combination bands of C—H stretching vibrations from –HC = CH–(*cis*) groups of unsaturated fatty acids and the combined spectrum band of C═O stretching vibration (Vieira, Assis, de Queiroz, Neves, & de Oliveira, 2021). In the visible wavelength range, the peaks at 416 nm, 456 nm, and 484 nm correspond to a variety of carotenoids. Moreover, a peak at 671 nm is correlated with chlorophyll and its derivatives (Tarhan, 2020). As can be seen from Fig. 1B, the main spectral differences are attributed to carotenoids, chlorophyll, and their derivatives. With the increase of adulteration level, the absorbance in the range of 400–760 nm gradually decreases.

Fluorescence is also an optical phenomenon in which the molecular absorption of energy in the form of photons triggers the emission of fluorescent photons with a longer wavelength. Generally, substances that produce fluorescence should have a rigid conjugated system. There are not many molecules with rigid conjugate systems in nature, so not all of them can emit fluorescence. The main substances that can produce fluorescence in edible oil are vitamin E, phenolic compounds, pigment and so on. Fig. 2 shows the representative EEM spectrum of pure and adulterated olive oil samples, and illustrates the changes observed from 100 % olive oil (left side) to 50 % olive oil. Two fluorescent zones were detected in pure and adulterated olive oil samples. In the first region, there are two fluorescent absorption peaks in the excitation wavelength range of 300–400 nm and emission wavelength range of 340–630 nm. And in the second region, there are five main fluorescent absorption peaks in the excitation wavelength range of 300–700 nm and emission wavelength range of 650–720 nm.

As noted in the current study, the short wavelength band, with excitation at 270–330 nm and emission wavelength at 280–330 nm, has been attributed to phenolic compounds (Omwange et al., 2021). The strong characteristic band is associated with tocopherol and tocotrienol with excitation at 300–360 nm and emission at 350–400 nm (Martín-Tornero et al., 2022). Low-intensity doublet bands exist in the excitation at 368–420 nm and emission at 430–470 nm. It has been to have a well-known relationship with conjugated hydroperoxides and oxidation products of unsaturated fatty acids (Dupuy, Le Dreau, Ollivier, Artaud, Pinatel, & Kister, 2005). In general, due to the high content of poly-unsaturated fatty acids in soybean oil, it is easy to oxidize to form conjugated hydroperoxides. As the concentration of the adulterating soybean oil increases, the EEM fluorescence intensity in this region increases. A band appears near 350 nm in excitation and 520 nm in emission, may be assigned to tocopherol and β-carotene absorptions (Magalhaes, Caires, Silva, Alcantara, & Oliveira, 2014; Zandomeneghi, Carbonaro, & Zandomeneghi, 2006). In the second region, olive oil presents five low-intensity bands, with excitation at 320–680 nm and emission at 650–700 nm. And these bands correspond to pheophytin and chlorophyll derivatives (Uncu & Ozen, 2019). In summary, the fatty acid oxidation products increase and chlorophyll content decrease with the increase of adulterating soybean oil concentration. As can be observed, there is an increase of the band intensity at the excitation of 330–440 nm and the emission of 660–700 nm, while a decrease of the intensity is observed in the range with excitation at 320–680 nm and emission at 650–700 nm.

Olive oil and soybean oil have similar chemical components. As a result, pure olive oil and adulterated olive oil samples have similar spectral spectrums, which are difficult to distinguish visually. Thus, it is necessary to analyze the differences between spectra by chemometric methods to detect the adulteration of olive oil.

3.2. Discrimination of olive oils from adulterated oils based on PCA and M-PCA

The PCA model was initially constructed on spectra datasets of samples, and the cluster analysis was performed on pure olive oil samples and their different adulterants with soybean oil. Prior to the model development, pre-processing of raw spectral data is indispensable in applying chemometrics when the measured spectra are affected by significant noises, baselines and other undesirable factors. There are many pre-processing methods used for the transformation of spectra data. The most frequently used methods include smoothing, derivative, multiplicative scatter (or signal) correction (MSC), standard normal variate (SNV), orthogonal signal correction (OSC), Savitzky-Golay (SG), and wavelet denoising techniques (WDTs) (Xu et al., 2008). Spectra pre-processing in chemometrics is still trial and error. In this work, model attempts are mainly adopted to select the best preprocessing method suitable for each spectroscopic method. In this study, for FTIR and NIR spectra, we used the full spectrum data. For the fluorescence spectra, in order to reduce the amount of data calculation and eliminate unnecessary interference, we removed the area with almost no fluorescence and retained the most fluorescence information as much as possible. The spectral range for fluorescence data analysis is 298–688 nm for excitation wavelength and 350–708 nm for emission wavelength.

First, for the FTIR spectra, five data pretreatment methods (mean centering, data normalization, SNV, first derivative, and second derivative) were applied to the PCA model to compare the results of clustering trend before data analysis. The comparative study results showed that the PCA model of FTIR with first derivative pretreatment was satisfactory in separating olive oil from adulterated oils. The result of the PCA model showed that PC1 explained 65.37 % of the total variance in the data set while PC2 explained 19.38 %. The cumulative percent variance (CPV) of the first three principal components was 84.75 %, which met the requirements of PCA analysis (CPV > 70-85 %) ((Martens & Naes, 1992; Liu, 2007). Therefore, it is believed that the first two principal components represent most of the original data information, which can be used to draw the analysis result graph.

Fig. 3A shows the PCA result for FTIR datasets scores plots of adulterated olive oil samples. It can be seen that each type of sample has a tendency to cluster with each other, and the pure olive oils and with different levels of adulterated olive oil can be separated basically. But one 5 % adulterated olive oil sample and four 30 % adulterated oil samples do not cluster with other oils at these adulterated levels.

For the Vis-NIR spectra dataset, the results showed that the PCA model with SNV as pretreatment had a better separation effect on separating olive oil from adulterated oils. The PCA showed that the total variance of PC1 and PC2 explained 99.7 % of the variance contribution rate in the data set. Thus, the PCA score in Fig. 3B shows the plotting results of Vis-NIR spectral data of adulterated olive oil samples. It can be seen that the olive oil samples with different levels of adulteration have

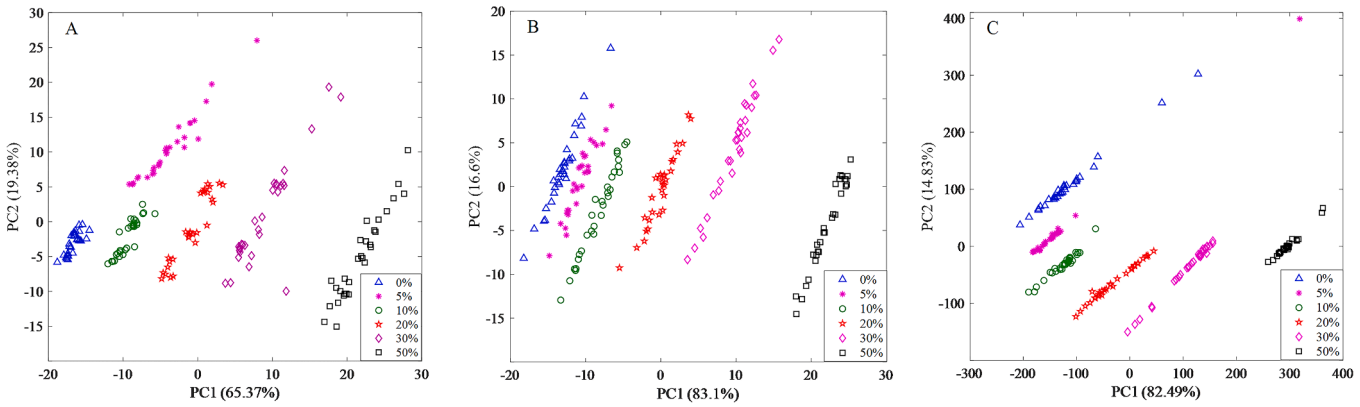


Fig. 3. The M−PCA for FTIR (A), Vis-NIR (B) and EEMs (C) datasets scores plots of adulterated olive oil samples.

some cluster tendency. However, the results are still not perfect. The olive oil samples adulterated with soybean oil at ratios of 20 %, 30 %, and 50 % can be distinguished well from each other, while 5 % and 10 % adulterated olive oil samples are too close to pure olive oil samples.

Fluorescence detection of the adulterated olive oil samples in this work yielded three-dimensional datasets. M-PCA was used to analyze the EEMs dataset. The first derivative combined with mean centering pretreatment was performed to remove undesirable factors from the raw spectral data before chemometrics analysis. The total variance contribution rate of PC1 and PC2 was 97.32 %, which met the requirement of principal component analysis. Fig. 3C displays the M-PCA score diagram of adulterated olive oil data with PC1 and PC2. It can be seen that some adulterated oil samples are far away, and are not clustered with other oil samples in these clusters. However, it can still be found that olive oil with different adulteration levels presents a clustering trend.

In general, the PCA and M-PCA score diagram of the three types of spectra shows that the adulterated olive oil in each proportion has a clustering trend and is basically distinguishable from each other. The PCA results give a promising scenario for applying an unsupervised statistical technique. Subsequently, the supervised classification method will be used to study the classification of adulterated olive oil resamples.

3.3. Discrimination of olive oils from adulterated olive oils based on PLS-DA and U-PLS-DA

PLS-DA was used to resolve FTIR and Vis-NIR, while U-PLS-DA was used for analyzing EEM spectroscopy. For convenience, the variety membership of each type of oil sample was coded as a dummy vector by assigning a value of 1 if the spectrum belonged to a particular class or 0 if it did not belong to this class. Then the olive oil with different ratios of adulteration could be coded as: 0 % (1, 0, 0, 0, 0, 0), 5 % (0, 1, 0, 0, 0, 0), 10 % (0, 0, 1, 0, 0, 0), 20 % (0, 0, 0, 1, 0, 0), 30 % (0, 0, 0, 0, 1, 0), 50 % (0, 0, 0, 0, 0, 1). PLS is a supervised classification method. When constructing the PLS model, spectral datasets need to be divided into training sets and prediction sets. FTIR, Vis-NIR and EEMs datasets of this work are composed of 180 olive oil samples (including 30 pure olive oil samples). A certain number of sample data were randomly selected from each category to form the training sets, and the remaining samples to form the prediction sets. Finally, the training sets contained 123 samples, and the prediction sets contained 57 samples. To obtain the best classification results, PLS-DA/U-PLS-DA requires proper selection of the latent variables (LVs) to avoid overfitting or underfitting the model. All the spectral methods adopted full cross-validation method (also called leave-one-out cross-validation) to select the optimum latent variables number in this study (Riedl, Esslinger, & Fauhl-Hassek, 2015). Take FTIR as an example, the best classification effect can be achieved when the number of LVs is 13 after cross-validation. Fig. 4A reveals the estimated dummy vectors by PLS-DA for the training and prediction sets of oil samples. According to its dummy vectors, the following rules were used to restore the dummy vectors matrix to 0 and 1 code matrix and identify it on this basis. If the first value in the prediction dummy vectors was the largest, then the maximum value was assigned to 1, and the other values were assigned to 0 so that the dummy vectors were converted into a code (1, 0, 0, 0, 0, 0), determining it is 0 % adulteration, i.e. pure olive oil. Similarly, all dummy vectors in the database could be converted into coding moments. Via careful inspection of Fig. 4A, one can obtain the final classification results of the training and predicted sets. Table 1 lists the classification results of oil samples. As can be seen from Table 1, the total recognition accuracy of both the training set and the prediction set is 100 % for the FTIR spectral data based on PLS-DA.

Similar operations to FTIR were used for Vis-NIR spectral analysis. The best classification effect could be achieved when the number of LVs is 13 after cross-validation. Fig. 4B shows the estimated dummy vectors estimated by the PLS-DA model based on Vis-NIR spectral for the training and prediction set of oil samples. Table 1 reveals the classification results by PLS-DA for Vis-NIR spectra, from which it can be found that the discriminated result of both prediction sets and the training sets are all correct, and the total recognition rates for the training and prediction sets are 100 %.

For EEMs data, U-PLS-DA is applied. Similarly, 123 samples of the 180 samples were randomly selected as the training set and the remaining 57 samples as the prediction set. Leave-one-out cross-validation was used to select the optimum latent variables number. The estimated dummy vectors estimated by the U-PLS-DA model based on EEMs for the training and prediction set of oil samples are shown in Fig. 4C. The final classification result of the training and predicted sets is also shown in Table 1. It can be seen from Table 1 that the oil samples with different adulteration levels have different degrees of overlap. In the training sets, the discriminated result was correct, and the total recognition rate was 100 %. In the prediction set, 2 samples of 5 % adulterated olive oil were mistakenly judged as pure olive oil, 4 samples of 10 % adulterated olive oil were mistakenly judged as 5 % of adulterated olive oil, 6 samples of 20 % adulterated olive oil were mistakenly judged as 30 % of adulterated olive oil, 1 sample of 20 % adulterated olive oil sample was misjudged as 10 % olive oil, 1 sample of 30 % adulterated olive oil sample was misjudged as 20 % olive oil, and 1 sample of 50 % adulterated olive oil sample was misjudged as 20 % olive oil. The total correct recognition rate of the prediction set was only 73 %.

As can be seen from the previous discussion, PCA/M-PCA and PLS-DA/U-PLS-DA combined three spectral methods for olive oil adulteration research results show that the FTIR and Vis-NIR analysis are relatively better. The possible reason is that more information is available in FTIR and Vis-NIR analysis. Almost all organic compounds in olive oil and soybean oil can cause FTIR and NIR absorption. Ingredients not present in olive oil, but present in soybean oil will be reflected in the infrared and near-infrared spectra of adulterated olive oil. Conversely,

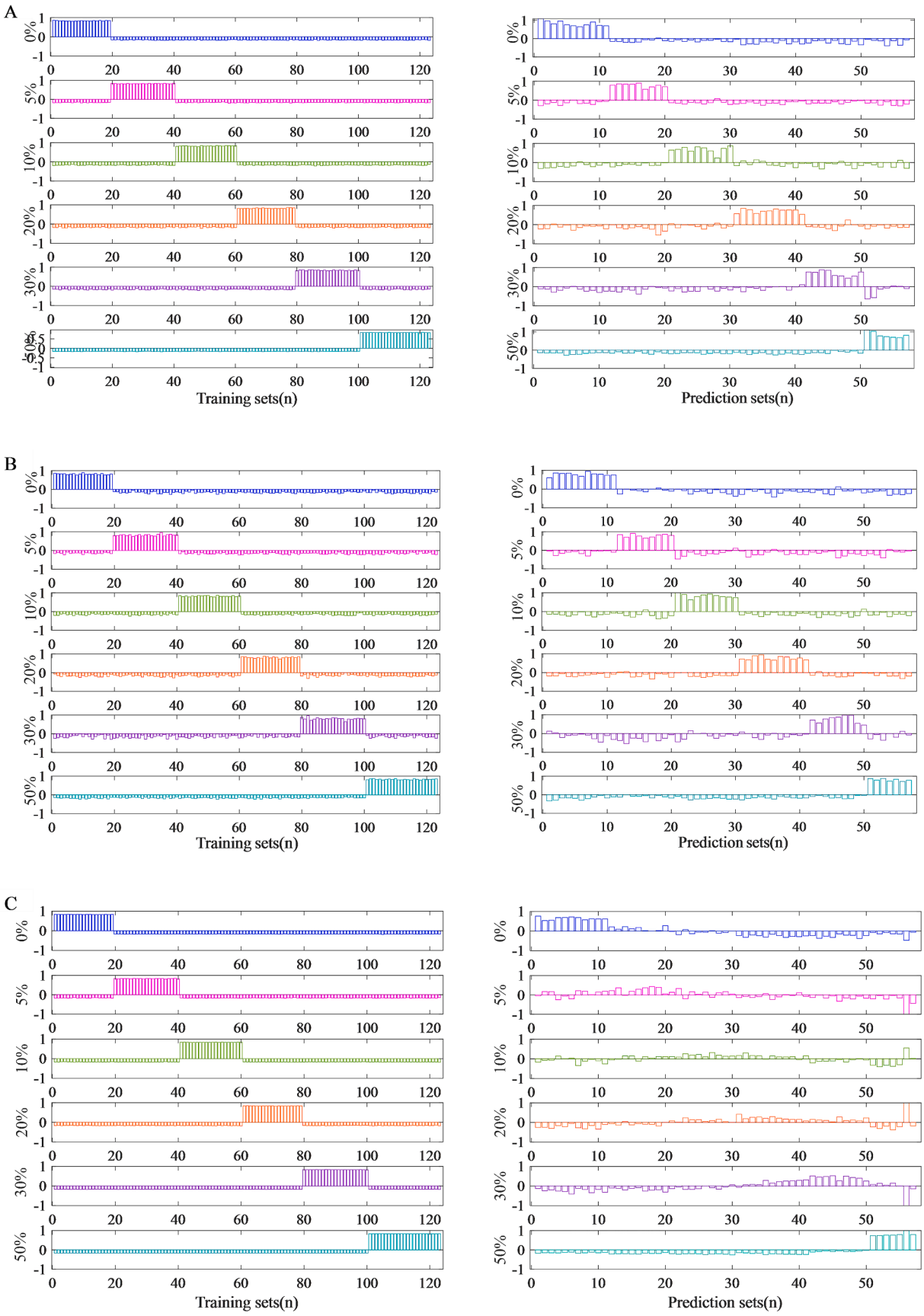


Fig. 4. The estimated dummy vectors by PLS-DA for FTIR (A) and Vis-NIR (B) while U-PLS-DA for EEMs (C) to the various memberships of olive oil samples in the training sets and the prediction sets.

Table 1
PLS-DA results of different spectral methods and SWATLD-BP-ANN results of fluorescence spectra of olive oil samples.

		Training sets (123)								Prediction sets (57)							
		Number	0 %	5 %	10 %	20 %	30 %	50 %	RR (%)	Number	0 %	5 %	10 %	20 %	30 %	50 %	RR (%)
FTIR-PLS-DA	0 %	19	19	0	0	0	0	0	100 %	11	11	0	0	0	0	0	100 %
	5 %	21	0	21	0	0	0	0	100 %	9	0	9	0	0	0	0	100 %
	10 %	20	0	0	20	0	0	0	100 %	10	0	0	10	0	0	0	100 %
	20 %	19	0	0	0	19	0	0	100 %	11	0	0	0	11	0	0	100 %
	30 %	21	0	0	0	0	21	0	100 %	9	0	0	0	0	9	0	100 %
	50 %	23	0	0	0	0	0	23	100 %	7	0	0	0	0	0	7	100 %
Vis-NIR-PLS-DA	0 %	19	19	0	0	0	0	0	100 %	11	11	0	0	0	0	0	100 %
	5 %	21	0	21	0	0	0	0	100 %	9	0	9	0	0	0	0	100 %
	10 %	20	0	0	20	0	0	0	100 %	10	0	0	10	0	0	0	100 %
	20 %	19	0	0	0	19	0	0	100 %	11	0	0	0	11	0	0	100 %
	30 %	21	0	0	0	0	21	0	100 %	9	0	0	0	0	9	0	100 %
	50 %	23	0	0	0	0	0	23	100 %	7	0	0	0	0	0	7	100 %
EEMs-U-PLS-DA	0 %	19	19	0	0	0	0	0	100 %	11	11	0	0	0	0	0	100 %
	5 %	21	0	21	0	0	0	0	100 %	9	2	7	0	0	0	0	78 %
	10 %	20	0	0	20	0	0	0	100 %	10	0	4	6	0	0	0	40 %
	20 %	19	0	0	0	19	0	0	100 %	11	0	0	1	4	6	0	37 %
	30 %	21	0	0	0	0	21	0	100 %	9	0	0	0	1	8	0	89 %
	50 %	23	0	0	0	0	0	23	100 %	7	0	0	0	1	0	6	86 %
EEMs-SWATLD-BP-ANN	0 %	19	19	0	0	0	0	0	100 %	11	11	0	0	0	0	0	100 %
	5 %	21	0	21	0	0	0	0	100 %	9	0	9	0	0	0	0	100 %
	10 %	20	0	0	20	0	0	0	100 %	10	0	0	10	0	0	0	100 %
	20 %	19	0	0	0	19	0	0	100 %	11	0	0	0	11	0	0	100 %
	30 %	21	0	0	0	0	21	0	100 %	9	0	0	0	0	9	0	100 %
	50 %	23	0	0	0	0	0	23	100 %	7	0	0	0	0	0	7	100 %

*RR denotes recognition rate.

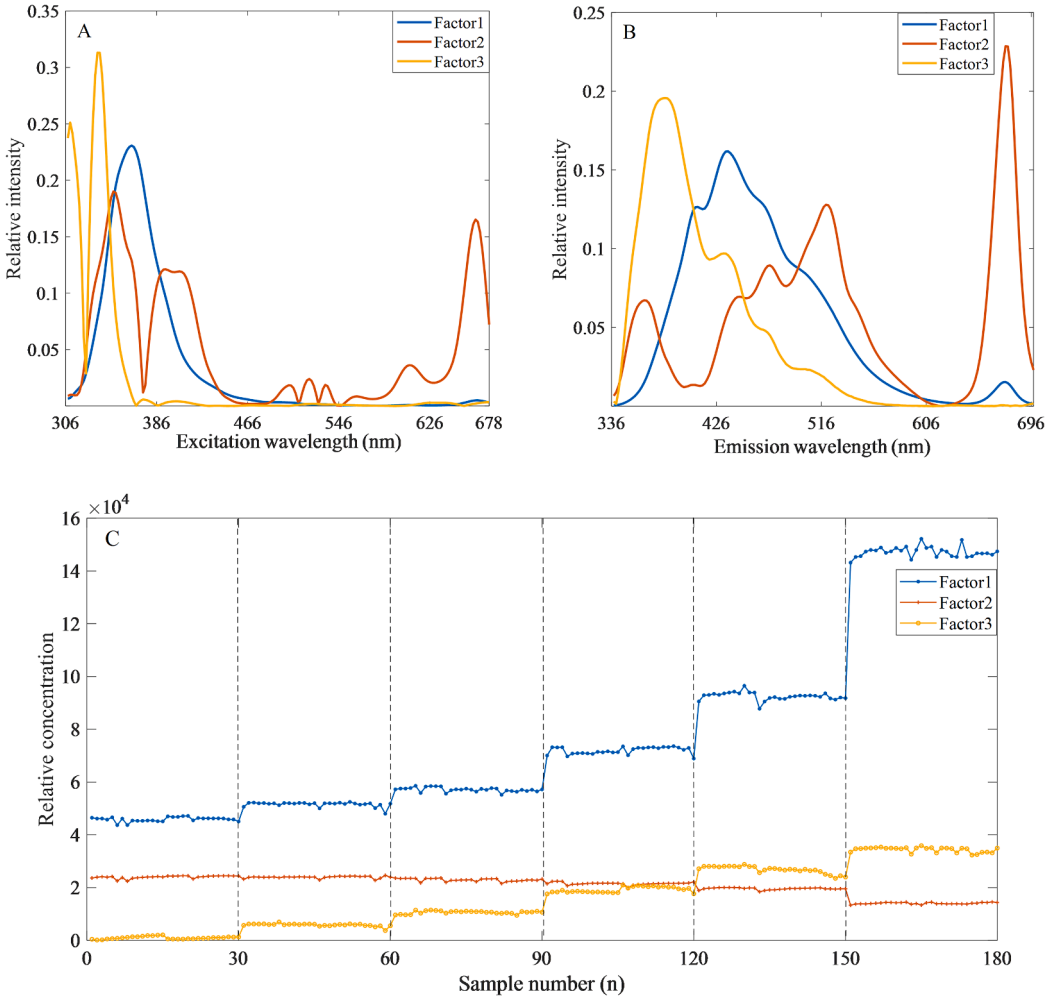


Fig. 5. SWATLD model concentration mode loadings: excitation profiles (A), emission profiles (B), and relative concentration (C).

edible oil has a few fluorescent substances and low levels of them. Thus, the amount of information that the fluorescence spectrum can provide is less. In other words, the difference in spectral information caused by fluorescent substances in adulterated olive oil is less than that caused by FTIR or Vis-NIR spectra. This may be the reason why the EEMs classification results are not as good as those of FTIR and Vis-NIR spectroscopy. It has been noted that EEMs combined with chemometrics have been reported in the literature for edible oil authenticity detection with desirable good results (Wu, Zhao, Tian, Shang, & Liu, 2020). Subsequently, we also tried some other chemometrics methods to analyze the fluorescence data to identify adulterated olive oil.

3.4. Swatld-BP-ANN

Considering that three-way data is not easy to be directly recognized by pattern recognition, we first decomposed EEMs data through a three-way decomposition algorithm and then conducted subsequent classification research by 2D pattern recognition method. In this work, SWATLD, a popular three-way decomposition algorithm, was chosen to decompose EEMs datasets. The EEMs datasets of 180 olive oil samples (pure and adulterated) were arranged in the three-way arrays of the dimensions as 180 × 216 × 216 (samples × number of emission × number of excitation). Decomposition of the arrays by SWATLD algorithm could obtain the spectral profiles (the loadings) and the relative concentrations (the scores) of components respectively. Just as LVs should be determined first before PLS, the chemical rank (i.e., the optimum number of factors in the system studied) should be determined before SWATLD decomposition to avoid underestimation or overestimation of the chemical rank resulting in subsequent model errors. Core consistency diagnosis (CORCONDIA) suggested that EEM datasets of oil samples requested to be fitted exactly with 3 factors. Therefore, the subsequent running of the SWATLD model used 3 factors. Meanwhile, non-negativity constraint was applied in all three modes when using the SWATLD model, since both the fluorescence intensities and concentrations were positive.

Fig. 5 shows the analysis results of the three-dimensional fluorescence data of the adulterated olive oil using 3 factors of the SWATLD decomposition model. Fig. 5A and B show the excitation spectra and the emission spectra of the 3 components in the adulterated olive oil analyzed by the SWATLD model, respectively. Fig. 5C shows the relative concentrations of three fluorophores in every measured oil sample. Many substances can generate fluorescence in vegetable oil, but there are only 3 substances in spectral data decomposed by SWATLD. The reason may be that the structures of these substances are too similar and they are classified into one category in the model analysis process. It is also possible that the content of some fluorescent substances in the oil samples selected is too small, and the fluorescence response intensity is too weak to be detected. At the same time, from the spectral wavelengths in Fig. 5A and B, it can be inferred that factor 1 and factor 3 with relatively obvious changes in relative concentrations may be related to tocopherols and the oxidation products of unsaturated fatty acids. As a polyunsaturated fatty acid, the content of linoleic acid in soybean oil is relatively high, and it is more easily oxidized. The concentration of its oxidation product is relatively high in soybean oil. The relative concentrations of factor 1 and factor 3 also increased with the increase of the concentration of the adulterating soybean oil. Meanwhile, it can be found that the relative concentration of component 3 in pure olive oil is the lowest, and the relative concentration is the highest in olive oil with an adulteration ratio of 50 %. Factor 2 may correspond to the fluorescence of pigments (chlorophyll, lutein, carotenoids, etc.) in olive oil and soybean oil. Since both olive oil and soybean oil contain pigments, when the proportion of adulteration is relatively small, this component did not change significantly, but when the adulteration ratio increased to 30 %, the chlorophyll content decreased, and the relative content of component 2 also decreased. In addition, it can be found from Fig. 5C that the relative concentration profiles of the three fluorescent substances in olive oil samples with different levels of adulteration are very different. However, there is no significant difference in the content of each of the fluorescence components in olive oil samples of the same adulteration level. And with the increase of the adulteration ratio, the relative concentrations of component 1 and component 2 in oil samples gradually increased. It was indicated that the difference in fluorescence relative concentration of oil samples with different adulteration levels was mainly caused by the incorporation of different quantities of soybean oil.

In spectral profiles and the relative concentrations decomposed by the SWATLD model, spectral profiles were similar and only the relative concentrations were significantly different. Subsequently, these relative concentration data were combined with the BP-ANN model to distinguish the adulteration of olive oil. In the BP-ANN model, 180 sets of data collected were randomly assigned, including 123 samples data for training sets and 57 samples data for prediction sets. A three-layer structure neural network was adopted, in which the number of neurons in the input layer was 3, the number of neurons in the hidden layer was 3, and the number of neurons in the output layer was 1. The maximal number of epochs was fixed at 20000, the learning rate was set to 0.5, and the target error was 0.01. Table 1 reveals the results of SWATLD-BP-ANN combined EEMs, from which it can be found that the correct recognition rate of both training sets and prediction sets is 100 %, indicating that the algorithm has better prediction and recognition effect. The model shows an optimal performance, being considered robust and reliable. It shows that the combination of the relative concentration decomposed by the three-way algorithm and 2D pattern recognition can distinguish adulterated olive oil from each other well. The possible reason is that the redundant variables in the original EEM spectrum are removed in the process of decomposition of the relative concentration mode by a three-way algorithm, and the discrimination result is superior to the U-PLS-DA discrimination result based on the original EEM spectral information. It should be noted that the calculation process based on a three-way algorithm becomes more complex and difficult to operate.

4. Conclusions

In this study, three common spectral methods (FTIR, Vis-NIR, and EEMs) combined with chemometrics were used to conduct comparative studies to distinguish adulterated olive oil. It shows that all three spectroscopy methods can obtain relatively satisfactory results on the detection of olive oil adulteration. Compared with EEMs, more satisfactory recognition results can be obtained from FTIR and Vis-NIR spectroscopy based on common pattern recognition methods such as PCA and PLS-DA. For FTIR and Vis-NIR spectra, the total recognition accuracy of PLS-DA for both the training set and the prediction set has reached 100 %. This is mainly because almost all the substances in olive oil and soybean oil can produce FTIR and Vis-NIR absorption spectrum, and relatively few of them can produce fluorescence. Therefore, FTIR and Vis-NIR spectra can provide more information than fluorescence spectra. After analyzing EEMS data using SWATLD, the BP-ANN algorithm can also perfectly identify olive oil adulterated with soybean oil. It shows 100 % accuracy. However, the whole process is more complicated to operate.

Compared to other analytical techniques, spectroscopy techniques are non-destructive, rapid, and do not require the pretreatment of samples. Meanwhile, combined with the pattern recognition method, satisfactory prediction results of adulterated olive oil can be obtained. In general, it is clear that the three spectroscopic techniques in combination with chemometrics are a promising tool in the prediction of adulterant oil in olive oil samples. Among the three spectroscopic analysis techniques, FTIR and Vis-NIR spectroscopy are more suitable for the rapid identification of adulterated olive oil because FTIR and Vis-NIR spectroscopy provide more information than fluorescence spectroscopy, and are relatively simple to operate and build model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

The work was financially supported by National Natural Science Foundation of China (Grant no. 21275039) and Natural Science Foundation of Henan Province, China (Grant no. 182102310681).

References

Amit, Jamwal, R., Kumari, S., Dhaulaniya, A. S., Balan, B., & Singh, D. K. (2020). Application of ATR-FTIR spectroscopy along with regression modelling for the detection of adulteration of virgin coconut oil with paraffin oil. *LWT-Food Science and Technology, 118*, Article 108754. https://doi.org/10.1016/j.lwt. 2019.108754

Bermudez, B., Lopez, S., Ortega, A., Varela, L. M., Pacheco, Y. M., Abia, R., & Muriana, F. J. G. (2011). Oleic acid in olive oil: From a metabolic framework toward a clinical perspective. *Current Pharmaceutical Design, 17*(8), 831–843. https://doi.org/10.2174/138161211795428957

Bucciantini, M., Leri, M., Nardiello, P., Casamenti, F., & Stefani, M. (2021). Olive polyphenols: Antioxidant and anti-inflammatory properties. *Antioxidants, 10*(7), 1044. https://doi.org/10.3390/antiox10071044

Carranco, N., Farres-Cebrian, M., Saurina, J., & Nunez, O. (2018). Authentication and quantitation of fraud in extra virgin olive oils based on HPLC-UV fingerprinting and multivariate calibration. *Foods, 7*(4), 44. https://doi.org/10.3390/foods7040044

Chen, Z. P., Wu, H. L., Jiang, J. H., Li, Y., & Yu, R. Q. (2000). A novel trilinear decomposition algorithm for second-order linear calibration. *Chemometrics and Intellgent Laboratory Systems, 52*(1), 75–86. https://doi.org/10.1016/S0169-7439(00)00081-2

Chen, Z. P., Wu, H. L., & Yu, R. Q. (2001). On the self-weighted alternating trilinear decomposition algorithm—the property of being insensitive to excess factors used in calculation. *Journal of Chemometrics, 15*(5), 439–453. https://doi. org/10.1002/cem.633.

da Silva, A. C., Soares, S. F. C., Insausti, M., Galvao, R. K. H., Band, B. S. F., & de Araujo, M. C. U. (2016). Two-dimensional linear discriminant analysis for classification of three-way chemical data. *Analytica Chimica Acta, 938*, 53–62. https://doi.org/10.1016/j.aca.2016.08.009

de Lima, T. K., Musso, M., & Bertoldo Menezes, D. (2020). Using Raman spectroscopy and an exponential equation approach to detect adulteration of olive oil with rapeseed and corn oil. *Food Chemistry, 333*, Article 127454. https://doi.org/10.1016/j. foodchem.2020.127454

Deng, Y., Zhou, X., Shen, J., Xiao, G., Hong, H. C., Lin, H. J., … Liao, B. Q. (2021). New methods based on back propagation (BP) and radial basis function (RBF) artificial neural networks (ANNs) for predicting the occurrence of haloketones in tap water [J]. *Science of The Total Environment, 772*, Article 145534. https://doi.org/10.1016/j.scitotenv.2021.145534

Du, Q. W., Zhu, M. T., Shi, T., Luo, X., Gan, B., Tang, L. J., & Chen, Y. (2021). Adulteration detection of corn oil, rapeseed oil and sunflower oil in camellia oil by in situ diffuse reflectance near-infrared spectroscopy and chemometrics. *Food Control, 121*, Article 107577. https://doi.org/10.1016/j.foodcont.2020.107577

Dupuy, N., Le Dreau, Y., Ollivier, D., Artaud, J., Pinatel, C., & Kister, J. (2005). Origin of French virgin olive oil registered designation of origins predicted by chemometric analysis of synchronous excitation-emission fluorescence spectra. *Journal of Agricultural and Food Chemistry, 53*(24), 9361–9368. https://doi.org/10.1021/jf051716m

Gendrin, C., Roggo, Y., & Collet, C. (2008). Pharmaceutical applications of vibrational chemical imaging and chemometrics: A review. *Journal of Pharmaceutical and Biomedical Analysis, 48*(3), 533–553. https://doi.org/10.1016/j.jpba.2008. 08.014

Harzalli, U., Rodrigues, N., Veloso, A. C. A., Dias, L. G., Pereira, J. A., Oueslati, S., & Peres, A. M. (2018). A taste sensor device for unmasking admixing of rancid or winey-vinegary olive oil to extra virgin olive oil. *Computers and Electronics in Agriculture, 144*, 222–231. https://doi.org/10.1016/j.compag.2017.12.016

Hou, X. W., Wang, G. L., Wang, X., Ge, X. M., Fan, Y. R., Jiang, R., & Nie, S. D. (2021). Rapid screening for hazelnut oil and high-oleic sunflower oil in extra virgin olive oil using low-field nuclear magnetic resonance relaxometry and machine learning. *Journal of the Science of Food and Agriculture, 101*(6), 2389–2397. https://doi.org/10.1002/jsfa.10862

Li, Y., Chen, S. Y., Chen, H., Guo, P., Li, T., & Xu, Q. (2020). Effect of thermal oxidation on detection of adulteration at low concentrations in extra virgin olive oil: Study based on laser-induced fluorescence spectroscopy combined with KPCA-LDA. *Food Chemistry, 309*, Article 125669. https://doi.org/10.1016/j.food chem.2019.125669

Liu, R. Q. (2007). *SPSS statistical software* (1st ed.). China Press of Traditional Chinese Medicine (Chapter 14).

Magalhaes, K. F., Caires, A. R. L., Silva, M. S., Alcantara, G. B., & Oliveira, S. L. (2014). Endogenous fluorescence of biodiesel and products thereof: Investigation of the molecules responsible for this effect. *Fuel, 119*, 120–128. https://doi.org/10.1016/j.fuel.2013.11.024

Martens, H., & Naes, T. (1992). *Multivariate calibration* ((2nd ed.)). John Wiley & Sons.

Martín-Tornero, E., Fernández, A., Pérez-Rodriguez, J. M., Durán-Merás, I., Prieto, M. H., & Martín-Vertedor, D. (2022). Non-destructive fluorescence spectroscopy as a tool for discriminating between olive oils according to agronomic practices and for assessing quality parameters. *Food Analytical Methods, 15*(2), 253–265. https://doi.org/10.1007/s12161-021-02112-2

Meras, I. D., Manzano, J. D., Rodriguez, D. A., & de la Pena, A. M. (2018). Detection and quantification of extra virgin olive oil adulteration by means of autofluorescence excitation-emission profiles combined with multi-way classification. *Talanta, 178*, 751–762. https://doi.org/10.1016/j.talanta.2017. 09.095

Monfreda, M., Gobbi, L., & Grippa, A. (2014). Blends of olive oil and seeds oils: Characterisation and olive oil quantification using fatty acids composition and chemometric tools. *Part II. Food Chemistry, 145*, 584–592. https://doi.org/10.1016/j.foodchem.2013.07.141

Omwange, K. A., Al Riza, D. F., Saito, Y., Suzuki, T., Ogawa, Y., Shiraga, K., … Kondo, N. (2021). Potential of front face fluorescence spectroscopy and fluorescence imaging in discriminating adulterated extra-virgin olive oil with virgin olive oil. *Food Control, 124*, Article 107906. https://doi.org/10.1016/j. foodcont.2021.107906

Pardo, R., Vega, M., Barrado, E., Castrillejo, Y., & Sanchez, I. (2013). Three-way principal component analysis as a tool to evaluate the chemical stability of metal bearing residues from wastewater treatment by the ferrite process. *Journal of Hazardous Materials, 262*(15), 71–82. https://doi.org/10.1016/j.jhazmat.2013. 08.031

Riedl, J., Esslinger, S., & Fauhl-Hassek, C. (2015). Review of validation and reporting of non-targeted fingerprinting approaches for food authentication. *Analytica Chimica Acta, 885*, 17–32. https://doi.org/10.1016/j.aca.2015.06.003

Rohman, A. (2017). The use of infrared spectroscopy in combination with chemometrics for quality control and authentication of edible fats and oils: A review. *Applied Spectroscopy Reviews, 52*(7), 589–604. https://doi.org/10.1080/05704928.2016.1266493

Romani, A., Ieri, F., Urciuoli, S., Noce, A., Marrone, G., Nediani, C., & Bernini, R. (2019). Health effects of phenolic compounds found in extra-virgin olive oil, by-products, and leaf of olea europaea l. *Nutrients, 11*(8). https://doi.org/10.3390/nu11081776

Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors[J]. *Nature, 323*(6088), 533–536.

Stark, A. H., & Madar, Z. J. N. (2002). Olive oil as a functional food: Epidemiology and nutritional approaches. *Nutrition Reviews, 60*(6), 170–176. https://doi.org/10.1301/002966402320243250

Tarhan, I. (2020). A comparative study of ATR-FTIR, UV-visible and fluorescence spectroscopy combined with chemometrics for quantification of squalene in extra virgin olive oils. *Spectrochimica Acta Part a-Molecular and Biomolecular Spectroscopy, 241*, Article 118714. https://doi.org/10.1016/j.saa.2020.118714

Uncu, O., & Ozen, B. (2020). Importance of some minor compounds in olive oil authenticity and quality. *Trends in Food Science & Technology, 100*, 164–176. https://doi.org/10.1016/j.tifs.2020.04.013

Uncu, O., & Ozen, B. (2019). A comparative study of mid-infrared, UV-Visible and fluorescence spectroscopy in combination with chemometrics for the detection of adulteration of fresh olive oils with old olive oils. *Food Control, 105*, 209–218. https://doi.org/10.1016/j.foodcont.2019.06.013

Vieira, L. S., Assis, C., de Queiroz, M. E. L. R., Neves, A. A., & de Oliveira, A. F. (2021). Building robust models for identification of adulteration in olive oil using FT-NIR, PLS-DA and variable selection. *Food Chemistry, 345*, Article 128866. https://doi.org/10.1016/j.foodchem.2020.128866

Vietina, M., Agrimonti, C., & Marmiroli, N. (2013). Detection of plant oil DNA using high resolution melting (HRM) post PCR analysis: A tool for disclosure of olive oil adulteration. *Food Chemistry, 141*(4), 3820–3826. https://doi.org/10.1016/j.foodchem.2013.06.075

Wu, X. J., Zhao, Z. L., Tian, R. L., Shang, Z. C., & Liu, H. L. (2020). Identification and quantification of counterfeit sesame oil by 3D fluorescence spectroscopy and convolutional neural network. *Food Chemistry, 311*, Article 125882. https://doi.org/10.1016/j.foodchem.2019.125882

Xu, L., Zhou, Y. P., Tang, L. J., Wu, H. L., Jiang, J. H., Shen, G. L., & Yu, R. Q. (2008). Ensemble preprocessing of near-infrared (NIR) spectra for multivariate calibration. *Analytica Chimica Acta, 616*(2), 138–143. https://doi.org/10.1016/j.aca.2008.04.031

Zandomeneghi, M., Carbonaro, L., & Zandomeneghi, G. (2006). Comment on: Excitation-emission fluorescence spectroscopy combined with three-way methods of analysis as a complementary technique for olive oil characterization. *Journal of Agricultural and Food Chemistry, 54*(14), 5214–5215. https://doi.org/10.1021/jf0605648

Zarezadeh, M. R., Aboonajmi, M., Varnamkhasti, M. G., & Azarikia, F. (2021). Olive oil classification and fraud detection using e-nose and ultrasonic system. *Food Analytical Methods, 14*(10), 2199–2210. https://doi.org/10.1007/s12161-021-02035-y